

TCS IPA 35 Marks Coding Revision Guide

This guide focuses only on the concepts most likely to appear in the 35-mark coding question. Practice the six questions in order. Each question introduces one new pattern while reinforcing previous concepts.

Quick Revision Sheet

Class, Constructor, Getters, Setters Arrays of Objects Scanner input Filtering objects Searching objects Finding maximum/minimum object HAS-A relationship Nested getter calls (obj.getA().getB()) Bubble Sort on objects Stable sorting (swap only when left<right) Return Object / Object[] / null Exact output formatting

Q1 Employee Filtering

Problem

Create Employee(id,name,department,salary). Implement: Employee[] findEmployees(Employee[] arr,double minSalary) Return employees whose salary >= minSalary, else null.

Solution Approach

Algorithm: 1.Read n employees. 2.Store in Employee[]. 3.Filter into temporary array. 4.Copy qualifying employees into result array. 5.Return null if count==0. Time: O(n).

Code Skeleton

Key Code Skeleton:

```
Employee[] temp=new Employee[arr.length];
int count=0;
for(Employee e:arr){
    if(e.getSalary()>=minSalary) temp[count++]=e;
}
if(count==0) return null;
Employee[] res=new Employee[count];
for(int i=0;i<count;i++) res[i]=temp[i];
return res;
```

Q2 Product Search + Maximum

Problem

Find the costliest product in a given category.

Solution Approach

Loop through array. Compare category and price. Return first maximum. O(n).

Code Skeleton

```
Product ans=null;
for(Product p:arr){
    if(p.getCategory().equalsIgnoreCase(cat)){
        if(ans==null||p.getPrice()>ans.getPrice())
            ans=p;
    }
}
return ans;
```

Q3 Hospital-Doctor (HAS-A)

Problem

Hospital has one Doctor. Filter hospitals by doctor's experience.

Solution Approach

Access nested object using hospital.getDoctor().getExperience().

Code Skeleton

```
if(h.getDoctor().getExperience()>=minExp){  
    ...  
}
```

Q4 Vehicle Sorting

Problem

Sort vehicles by price descending without Arrays.sort().

Solution Approach

Use Bubble Sort.

Code Skeleton

```
for(int i=0;i<arr.length;i++)  
    for(int j=0;j<arr.length-i;j++)  
        if(arr[j].getPrice()<arr[j+1].getPrice())  
            Vehicle t=arr[j];  
            arr[j]=arr[j+1];  
            arr[j+1]=t;  
}
```

Q5 University-Professor

Problem

HAS-A + Filter + Stable Bubble Sort.

Solution Approach

Filter first, then bubble sort by professor experience.

Code Skeleton

Compare:

```
u.getProfessor().getExperience()
```

Q6 Final Mock Hotel-Customer

Problem

Customer belongs to Hotel. Filter by spending and sort descending.

Solution Approach

This combines every IPA pattern.

Code Skeleton

Template:

Read Input

Create Objects

Store in Array

Call Method

Filter

Sort

Return

Print

Exam Checklist

Create classes correctly. Use constructors. Create object array. Read input carefully.

Write the required static method. Handle null and no-result cases. Use manual bubble sort if required. Print exactly in required format.